NIKHIL K

DevOps Engineer | CI/CD, Docker, Observability, Secure Infrastructure
Kannur, Kerala, India | +91 9567037727 | nikhil82647@gmail.com
linkedin.com/in/nikhil-k-7b564a2b9 | github.com/Nikhil-k00

PROFESSIONAL SUMMARY

Computer Science undergraduate (80.1%) and Hac'KP 2026 finalist (Kerala Police Cyberdome) with hands-on DevOps experience building CI/CD pipelines, containerized infrastructure and observability for a multi-service platform. Skilled in GitHub Actions, Docker Compose, Harbor, PostgreSQL, Keycloak, Nginx (HTTPS) and OpenTelemetry with Grafana. Also built AI systems in Python and Node.js.

TECHNICAL SKILLS

DevOps & CI/CD: GitHub Actions, CI/CD Pipelines, Docker, Docker Compose, Harbor Registry, SSH Deployment, Nginx, HTTPS/TLS, Linux, Git, GitHub
Observability & Security: OpenTelemetry, Grafana (LGTM stack), Keycloak (IAM), Least-Privilege Access Control
Databases: PostgreSQL, MySQL, MongoDB, SQL, Alembic
Programming & Frameworks: Python, Java, JavaScript, C, FastAPI, Node.js, Express.js, React.js
AI / ML: Machine Learning, Computer Vision, Agentic AI, LLMs, OpenCV, InsightFace

WORK EXPERIENCE

Hac'KP 2026 Finalist (DevOps) | Kerala Police Cyberdome | 2026
Agentic Child Protection Investigation Assistant (ACPIA), team-built cyber-investigation platform

- Designed CI/CD pipelines in GitHub Actions for multiple microservices: automated tests, Docker image builds, pushes to a private Harbor registry, SSH deployment and Discord notifications, including a uv-workspace monorepo with root-context build.
- Set up shared infrastructure with Docker Compose: PostgreSQL with least-privilege databases per service, Keycloak in production mode, and Nginx reverse proxies with HTTPS for each service domain.
- Rolled out observability with OpenTelemetry and the Grafana LGTM stack across traces, metrics and logs; resolved a duplicate-exporter conflict with FastAPI's built-in telemetry.
- Diagnosed production issues across teams (broken Alembic migration chains, port mismatches, container startup crash, missing Vite build-time settings) and worked with owning teams on fixes.
- Reviewed PRs for deployment readiness and defined the deploy format adopted for new services.

Tech: GitHub Actions, Docker, Docker Compose, Harbor, PostgreSQL, Keycloak, Nginx, OpenTelemetry, Grafana, FastAPI, uv, Alembic, Linux

AI Agent Architect Virtual Intern | CSRBOX - IBM SkillsBuild | Jul 2025 - Aug 2025, Remote

- Designed and implemented AI agents for real-world problem scenarios and built AI-powered workflows using Generative AI and Machine Learning concepts in a peer team.

PROJECTS

Exam Guard: AI-Powered Examination Authentication System | github.com/Nikhil-k00/Exam-Guard | Jan 2026 - Mar 2026

- Built a biometric authentication system with InsightFace and OpenCV; achieved over 95% face verification accuracy using embeddings and cosine similarity.
- Automated attendance and seat allocation with real-time facial recognition. **Tech:** Node.js, Express.js, MySQL, OpenCV, InsightFace

LumiLearn: Multi-Agent Adaptive Learning Platform for Dyslexic Children | github.com/Nikhil-k00/LumiLearn | Jul 2026 - Present

- Developed a multi-agent learning platform with agents for phonics, reading, writing and memory games, plus an orchestration agent that personalizes learning paths from student analytics.
- Built gamified activities and progress dashboards for students and parents. **Tech:** Python, LLMs, Node.js, Express.js, MongoDB, JavaScript. INCLUCODE Buildathon 2026 Finalist.

EDUCATION

B.Tech Computer Science and Engineering, Vimal Jyothi Engineering College, Kerala | Sep 2023 - Present | 80.1%
Higher Secondary (Bioscience), GHSS Kannadiparamba, Kannur | 2021 - 2023 | 90% SSLC, GHSS Kannadiparamba | 2021 | 98%
Coursework: Data Structures and Algorithms, Machine Learning, DBMS, Operating Systems, Computer Networks, Computer Organization and Architecture

CERTIFICATIONS

AI Agent Architect, IBM SkillsBuild (2025) | Docker Essentials, Cognitive Class (2025) | Introduction to AI, Google (2026) | SQL for Analyst, Google Developer Community (2025) | Web Development, IEEE Computer Society (2025) | Programming in C, Infosys Springboard (2024)

ACHIEVEMENTS AND VOLUNTEERING

- Hac'KP 2026 (Kerala Police Cyberdome) finalist; INCLUCODE Buildathon 2026 finalist for LumiLearn.
- National Service Scheme (NSS) Volunteer: organized environmental and public welfare programs. **Languages:** English, Malayalam